

Disproof of Conjecture 00000001007: the Suzuki–Tits ovoid of $Q(4, 8)$ is not an elliptic quadric

jilint777

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Abstract

Conjecture 00000001007 asserts that in even characteristic every ovoid of the parabolic quadric $Q(4, q)$ is an elliptic quadric. This is false for $q = 8$. The Suzuki–Tits ovoid (Tits, 1962), lifted to $Q(4, 8)$, is an ovoid: it has $q^2 + 1 = 65$ points and meets every line of $Q(4, 8)$ exactly once. But it lies in no hyperplane of $\text{PG}(4, 8)$, so it is not a hyperplane section $Q^-(3, 8)$ of $Q(4, 8)$. Its projection from the nucleus to $\text{PG}(3, 8)$ lies on no quadric at all, so it is not an elliptic quadric in that model either. All facts are checked in Lean 4 (core library only) and by an independent Python script. Short hand proofs are also given.

1 The conjecture and what we refute

The statement reads:

Conjecture: In even characteristic, every ovoid of $Q(4, q)$ is an elliptic quadric (uniqueness; the odd-characteristic case has Suzuki’s counterexamples) — the even-characteristic uniqueness conjecture.

The Chinese version says the same thing. The clause we refute is

(C) *for every even prime power q , every ovoid of $Q(4, q)$ is an elliptic quadric.*

One even value of q with a non-elliptic ovoid is enough, and we use $q = 8$. The parenthetical remark is not a mathematical claim we need. It is also inaccurate: the Suzuki–Tits ovoids exist only in characteristic 2, for $q = 2^{2e+1} \geq 8$. They are exactly the counterexamples to (C).

Readings. “Ovoid of $Q(4, q)$ ” has a standard meaning (Definition 1). We also use the standard equivalent finite form (Lemma 2). “Elliptic quadric” can be read in two models.

- (R1) *Inside* $\text{PG}(4, q)$. An elliptic quadric ovoid of $Q(4, q)$ is a section $Q(4, q) \cap H$ by a hyperplane H such that $Q|_H$ is elliptic ($Q^-(3, q)$). The same holds for any image of such a section under a collineation or a semilinear isometry, since these maps send hyperplanes to hyperplanes. Every such set lies in a hyperplane of $\text{PG}(4, q)$.
- (R2) *Inside* $\text{PG}(3, q)$. For q even, projection from the nucleus N of $Q(4, q)$ is a bijection from the points of $Q(4, q)$ onto the points of $\text{PG}(3, q)$, and ovoids of $Q(4, q)$ map to ovoids of $\text{PG}(3, q)$. In this model “elliptic quadric” means the zero set of an elliptic quadratic form on $\mathbb{F}_q^4$, possibly after a collineation. Every such set lies on a quadric, i.e. on the zero set of a nonzero quadratic form, since collineations map quadrics to quadrics.

We exhibit an ovoid $\mathcal{T}$ of $Q(4, 8)$ that lies in no hyperplane and whose projection lies on no quadric. This defeats both readings and any reading that implies one of these two necessary conditions.

2 The objects

The field. $\mathbb{F}_8 = \mathbb{F}_2[\omega]/(\omega^3 + \omega + 1)$. We code $b_0 + b_1\omega + b_2\omega^2$ by the integer $b_0 + 2b_1 + 4b_2$. So $2 = \omega$, $4 = \omega^2$, $3 = \omega^3$, $6 = \omega^4$, $7 = \omega^5$, $5 = \omega^6$, and ω has order 7. Addition is XOR of codes, and $\text{char } \mathbb{F}_8 = 2$, so $q = 8$ is even. The map $\sigma(x) = x^4$ is a field automorphism with $\sigma(\sigma(x)) = x^{16} = x^2$, so σ^2 is the Frobenius map. This is the automorphism needed for the Tits ovoid when $q = 2^{2e+1}$ with $e = 1$.

The quadric. On $V = \mathbb{F}_8^5$ let

$$Q(x) = x_0x_3 + x_1x_2 + x_4^2, \quad f(x, y) = x_0y_3 + x_3y_0 + x_1y_2 + x_2y_1.$$

The identities $(a + b)(c + d) = ac + bd + (ad + bc)$ and $(a + b)^2 = a^2 + b^2$ give $Q(x + y) = Q(x) + Q(y) + f(x, y)$, so f is the polar form of Q . Also $f(x, x) = 2(x_0x_3 + x_1x_2) = 0$. Pairing with the unit vectors e_3, e_2, e_1, e_0 gives $f(x, \cdot) = 0$ if and only if $x_0 = x_1 = x_2 = x_3 = 0$. So the radical of f is spanned by the nucleus $N = (0, 0, 0, 0, 1)$, and $Q(N) = 1 \neq 0$. Hence Q is a nondegenerate quadratic form of parabolic type, and its zero set $Q(4, 8)$ in $\text{PG}(4, 8)$ is the parabolic quadric. All nondegenerate quadrics of $\text{PG}(4, q)$ are projectively equivalent, so the choice of model loses nothing: an isometry maps ovoids to ovoids and hyperplane sections to hyperplane sections.

For points $x \neq y$ of $Q(4, 8)$, $Q(sx + ty) = s^2Q(x) + t^2Q(y) + stf(x, y) = stf(x, y)$. So the line xy lies on the quadric if and only if $f(x, y) = 0$. Such points are called *collinear*. The lines of $Q(4, 8)$ are the totally singular 2-dimensional subspaces of V .

Counts. $Q(4, q)$ is a generalized quadrangle of order (q, q) [2, §3.1]. It has $(q + 1)(q^2 + 1)$ points and $(q + 1)(q^2 + 1)$ lines. Each line has $q + 1$ points and each point lies on $q + 1$ lines. For $q = 8$ there are 585 points, 585 lines, 9 points per line and 9 lines per point. `verify.py` confirms these numbers by listing all 4681 points of $\text{PG}(4, 8)$ and all lines of the quadric.

Definition 1. An *ovoid* of $Q(4, q)$ is a set $\mathcal{O}$ of points of $Q(4, q)$ that meets every line of $Q(4, q)$ in exactly one point.

Lemma 2. A set $\mathcal{O}$ of points of $Q(4, q)$ is an ovoid if and only if $|\mathcal{O}| = q^2 + 1$ and no two points of $\mathcal{O}$ are collinear.

Proof. ($\Leftarrow$) Two points of $\mathcal{O}$ on a common line of $Q(4, q)$ would be collinear, so each line contains at most one point of $\mathcal{O}$. Each point of $\mathcal{O}$ lies on $q + 1$ lines, and these lines are distinct for distinct points of $\mathcal{O}$. So $\mathcal{O}$ meets at least $(q^2 + 1)(q + 1)$ lines, which is all of them. ($\Rightarrow$) Two collinear points of $\mathcal{O}$ would put two points of $\mathcal{O}$ on one line. Counting incidences between $\mathcal{O}$ and lines gives $|\mathcal{O}|(q + 1) = (q + 1)(q^2 + 1)$. $\square$

Lean formalizes the right-hand side as `IsOvoid`, for $q = 8$. It also proves the “at most one point per line” half of ($\Leftarrow$) for every totally singular line (`IsOvoid.at_most_one`). The counting half is the argument above. `verify.py` checks the defining property directly: each of the 585 lines meets our set in exactly one point.

3 The Suzuki–Tits ovoid

In $\text{PG}(3, 8)$ the Tits ovoid [1] is

$$\mathcal{T}_3 = \{(1, x, y, xy + x^{\sigma+2} + y^\sigma) : x, y \in \mathbb{F}_8\} \cup \{(0, 0, 0, 1)\}, \quad x^{\sigma+2} = x^6, \quad y^\sigma = y^4.$$

We lift it to $Q(4, 8)$. Every $a \in \mathbb{F}_8$ has a unique square root, because squaring is an automorphism. We append to (p_0, p_1, p_2, p_3) the coordinate $\sqrt{p_0 p_3 + p_1 p_2}$. For an affine point this square root is $\sqrt{x^6 + y^4} = x^3 + y^2$. So

$$\mathcal{T} = \{t(x, y) := (1, x, y, xy + x^6 + y^4, x^3 + y^2) : x, y \in \mathbb{F}_8\} \cup \{(0, 0, 0, 1, 0)\}.$$

The point $t(x, y)$ is on the quadric because $Q(t(x, y)) = (xy + x^6 + y^4) + xy + (x^3 + y^2)^2 = 0$. The last point is on the quadric because $Q(0, 0, 0, 1, 0) = 0$. Projection from N (deleting x_4) maps $\mathcal{T}$ onto $\mathcal{T}_3$.

Proposition 3. *$\mathcal{T}$ is an ovoid of $Q(4, 8)$. It consists of 65 distinct points, no two of them collinear.*

Proof. The 64 points $t(x, y)$ have distinct coordinates (x_1, x_2) and first coordinate 1. The point $(0, 0, 0, 1, 0)$ is the only one with $x_0 = 0$. So the 65 points are pairwise non-proportional. Moreover $f((0, 0, 0, 1, 0), t(x, y)) = 1$, and for $(x, y) \neq (x', y')$,

$$f(t(x, y), t(x', y')) = (x + x')(y + y') + x^6 + x'^6 + (y + y')^4.$$

That this is nonzero is Tits' theorem [1]. Here it is checked exhaustively for all 2080 pairs, by **decide** in Lean and in **verify.py**. By Lemma 2, $\mathcal{T}$ is an ovoid. **verify.py** also checks Definition 1 directly on all 585 lines, and checks that $\mathcal{T}_3$ has no three collinear points (all 43680 triples), so $\mathcal{T}_3$ is an ovoid of $\text{PG}(3, 8)$. $\square$

4 $\mathcal{T}$ is not an elliptic quadric

We use one elementary fact: a polynomial over $\mathbb{F}_8$ of degree at most 7 that vanishes at all 8 elements of $\mathbb{F}_8$ is the zero polynomial.

Proposition 4 (reading R1). *No hyperplane of $\text{PG}(4, 8)$ contains $\mathcal{T}$.*

Proof. Let $a \in \mathbb{F}_8^5$ satisfy $\sum_i a_i p_i = 0$ for all $p \in \mathcal{T}$. The points $t(0, y) = (1, 0, y, y^4, y^2)$ give $a_0 + a_2 y + a_4 y^2 + a_3 y^4 = 0$ for all $y \in \mathbb{F}_8$. This polynomial has degree at most 4, so $a_0 = a_2 = a_3 = a_4 = 0$. Then $t(1, 0) = (1, 1, 0, 1, 1)$ gives $a_1 = 0$. So $a = 0$. $\square$

Proposition 5 (reading R2). *No nonzero quadratic form $c(x) = \sum_{0 \leq i \leq j \leq 3} c_{ij} x_i x_j$ on $\mathbb{F}_8^4$ vanishes on $\mathcal{T}_3$. Hence $\mathcal{T}_3$ lies on no quadric of $\text{PG}(3, 8)$.*

Proof. The point $(0, 0, 0, 1)$ gives $c_{33} = 0$. The points $(1, 0, y, y^4)$ give $c_{00} + c_{02} y + c_{22} y^2 + c_{03} y^4 + c_{23} y^5 = 0$ for all y . This polynomial has degree at most 5, so these five coefficients vanish. The points $(1, x, 0, x^6)$ now give $c_{01} x + c_{11} x^2 + c_{13} x^7 = 0$ for all x , since the terms with c_{00}, c_{03}, c_{33} are already zero. This polynomial has degree at most 7, so $c_{01} = c_{11} = c_{13} = 0$. Finally the point $(1, 1, 1, 1) = t(1, 1)$ with the last coordinate deleted gives $c_{12} = 0$. $\square$

Theorem 6. *Clause (C) is false: $\mathcal{T}$ is an ovoid of $Q(4, 8)$, $q = 8$ is even, and $\mathcal{T}$ is not an elliptic quadric under reading R1 or R2.*

Proof. Proposition 3 shows that $\mathcal{T}$ is an ovoid. An elliptic quadric under R1 lies in a hyperplane, which Proposition 4 rules out. Under R2 its projection lies on a quadric, which Proposition 5 rules out. $\square$

Control (the predicates are not vacuous). The polynomial $x^2 + x + 1$ has no root in $\mathbb{F}_8$. So the hyperplane $H : x_4 = x_1 + x_2$ cuts $Q(4, 8)$ in the elliptic quadric $x_0x_3 + x_1^2 + x_1x_2 + x_2^2 = 0$. That section is $\mathcal{E} = \{(1, a, b, a^2 + ab + b^2, a + b)\} \cup \{(0, 0, 0, 1, 0)\}$. Lean proves that $\mathcal{E}$ satisfies `IsOvoid`, lies in H , and lies on the quadric $x_0x_3 + x_1^2 + x_1x_2 + x_2^2$ of $\text{PG}(3, 8)$. `verify.py` checks that $\mathcal{E} = Q(4, 8) \cap H$ and that it meets every line once. $\mathcal{T}$ and $\mathcal{E}$ share only 5 points.

Remark 7. For $q = 2$ and $q = 4$ every ovoid of $\text{PG}(3, q)$ is an elliptic quadric (classical results of Barlotti and others), and hence so is every ovoid of $Q(4, q)$. So $q = 8$ is the smallest counterexample. For every $q = 2^{2^e+1} \geq 8$ the Tits ovoid gives a counterexample in the same way. For odd prime q , every ovoid of $Q(4, q)$ is an elliptic quadric (Ball, Govaerts and Storme). None of this is used above.

5 Lean formalization

The project `lean4/` uses Lean 4.19.0 and the core library only. It contains no `sorry`, no `native.decide` and no added axioms.

- **F8:** the eight elements, with addition and multiplication tables generated from $\omega^3 = \omega + 1$. The theorems `add_assoc`, `mul_inv`, `left_distrib` and so on check the field axioms exhaustively. `char_two` shows that the characteristic is 2, and `omega_generates` shows that ω has order 7. `sigma_props` shows that $\sigma(x) = x^4$ is an automorphism with $\sigma^2 = \text{Frobenius}$.
- **Q** and **pol** are Q and f . The theorem `Q_vadd` is the polarization identity, proved by algebra. `Q_nondegenerate` shows that the radical is $\langle N \rangle$, that $Q(N) = 1$, and that no nonzero singular vector lies in the radical.
- **IsOvoid 0:** O has 65 distinct entries, all nonzero and on Q , and $f(p, p') \neq 0$ for distinct entries. `IsOvoid.proj_distinct` shows that the entries are distinct projective points. `IsOvoid.at_most_one` shows that every totally singular line $\{sx + ty\}$ contains at most one entry.
- **tits** is $\mathcal{T}$, written with `sigma`. `tits_lift` shows that $(x^3 + y^2)^2 = x^{\sigma+2} + y^\sigma$, and `tits_isOvoid` is proved by `decide +kernel`.
- **InHyperplane 0** and **OnQuadricPG3 0** are the necessary conditions of **R1** and **R2**. `tits_not_inHyperplane` and `tits_not_onQuadric` prove that $\mathcal{T}$ fails both. These proofs do not enumerate the 8^5 or 8^{10} coefficient vectors. They use a general lemma (`dotL_comb, kill`): if c is orthogonal to the rows of E and $NE = I$, then $c = 0$. They also use explicit inverse matrices N for 5 points (linear forms) and 10 points (quadratic monomials) of $\mathcal{T}$, checked by `decide`.
- `conjecture_00000001007_false:` $\neg(\forall 0, \text{IsOvoid } 0 \rightarrow \text{InHyperplane } 0)$.
`conjecture_00000001007_false_any_reading:` for every predicate `IsElliptic` whose instances satisfy `InHyperplane` or `OnQuadricPG3`, $\neg(\forall 0, \text{IsOvoid } 0 \rightarrow \text{IsElliptic } 0)$.
`nonvacuous` and `ell_isOvoid` show that $\mathcal{E}$ satisfies all of these predicates.

Limitations. Lean fixes $q = 8$ and the standard model of $Q(4, 8)$; it does not formalize general finite fields. This is enough, because (C) quantifies over all even q . The ovoid predicate in Lean is the finite form of Lemma 2. Lean proves the “at most one point per line” half of that lemma. The counting half and the generalized-quadrangle counts are proved above, and `verify.py` checks the

line definition exhaustively. “Elliptic quadric” enters Lean only through its necessary conditions. Refuting those weaker conditions gives a stronger disproof.

6 Reproduction

```
cd lean4 && lake build      # about 40 s
cd .. && python3 verify.py   # about 10 s, standard library only
```

References

- [1] J. Tits, *Ovoïdes et groupes de Suzuki*, Arch. Math. 13 (1962), 187–198.
- [2] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, 2nd ed., EMS, 2009.